Home Assignment 1

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Exercise 1: Limits to Diversification

Part 1: Independent and identically distributed returns

We consider an equally weighted portfolio consisting of N securities with independent and identically distributed returns and finite variance.

Portfolio variance

$$\text{Var}(R_p) = \sigma_p^2 = \mathbf{w}'\Sigma\mathbf{w}$$

Equally portfolio weights:

$$w_i = \frac{1}{N}$$

Portfolio return

$$R_p = \sum_{i=1}^{N} w_i R_i$$

Substituting the equal portfolio weights:

$$R_p = \frac{1}{N} \sum_{i=1}^{N} R_i$$

Derivation of portfolio variance

Variance of the portfolio return:

$$\text{Var}(R_p) = \text{Var}\left(\frac{1}{N} \sum_{i=1}^{N} R_i \right)$$

Variance rule for linear combination:

$$\text{Var}(aX + c) = a^2 \, \text{Var}(X)$$

In our case.

$$\mathrm{Var}(R_p) = \frac{1}{N^2}\,\mathrm{Var}\left(\sum_{i=1}^{N} R_i\right)$$

Since the securities have independent returns, the variance of their sum equals the sum of their individual variances:

$$\mathrm{Var}(R_p) = \frac{1}{N^2}\sum_{i=1}^{N}\mathrm{Var}(R_i)$$

Because the returns are identically distributed, each security has the same variance:

$$\mathrm{Var}(R_i) = \sigma^2$$

Therefore:

$$\mathrm{Var}(R_p) = \frac{1}{N^2}\sum_{i=1}^{N}\sigma^2 = \frac{\sigma^2}{N}$$

Limit as N approaches infinity

Portfolio variance approaches zero as N increases:

$$\lim_{N\to\infty}\mathrm{Var}(R_p) = \lim_{N\to\infty}\frac{\sigma^2}{N} = 0$$

Part 2: Positive average covariance

Unlike Part 1, the securities are no longer independent.

For an equally weighted portfolio, the variance can be written as:

$$\sigma_p^2 = \frac{\sigma^2}{N} + \frac{N-1}{N}c$$

Here, σ^2 represents the average individual variance and c the average covariance between different securities.

As N approaches infinity:

$$\lim_{N\to\infty}\sigma_p^2 = 0 + c = c > 0$$

Therefore, portfolio variance does not approach zero because the positive average covariance remains.

Therefore, answer a is correct, as both parts have been proven.

Exercise 2: Expected Utility

The investment has two possible returns: -80% with a probability of 10%, and +15% with a probability of 90%.

The utility function is given in the assignment:

$$U(R) = E[R] - \frac{1}{2}\lambda \operatorname{Var}(R)$$

Results of my R code:

$$E[R] = 0.055$$

$$\operatorname{Var}(R) = 0.081225$$

$$U(R) = -0.026225$$

Therefore, answer c) is correct, since $-1 < U(R) < 0$.

R Code

```r
# Exercise 2: Expected Utility

# Given values
R <- c(-0.80, 0.15)
p <- c(0.10, 0.90)
lambda <- 2

# Expected return
expected_return <- sum(p * R)

# Variance
variance <- sum(p * (R - expected_return)^2)

# Utility
utility <- expected_return - 0.5 * lambda * variance

# Results
expected_return
variance
utility
```

Exercise 3: Insurance and Risk Aversion

Since the investor is indifferent between buying insurance and remaining uninsured, both alternatives must provide the same utility.

Using the utility function from Exercise 2:

$$U(R) = E[R] - \frac{1}{2}\lambda \operatorname{Var}(R)$$

Rearranging for λ:

$$\lambda = \frac{2(\mu_{\text{uninsured}} - \mu_{\text{insured}})}{\sigma^2_{\text{uninsured}} - \sigma^2_{\text{insured}}}$$

Using the values calculated in R:

$$\lambda = -1.7676768$$

Therefore, answer a is correct, since $\lambda < 0.5$.

R Code

```r
# Exercise 3: Insurance and Risk Aversion

# Given values
x <- 100000
premium <- 1000

# Returns without insurance
R_uninsured <- R

# Returns with insurance
R_insured <- c(-premium / x, 0.15 - premium / x)

# Expected returns
er_uninsured <- sum(p * R_uninsured)
er_insured <- sum(p * R_insured)

# Variances
var_uninsured <- sum(p * (R_uninsured - er_uninsured)^2)
var_insured <- sum(p * (R_insured - er_insured)^2)

# Indifference condition
lambda_3 <- 2 * (er_uninsured - er_insured) /
  (var_uninsured - var_insured)

lambda_3
```

Exercise 4: Matrix Calculations

First, the continuously compounded monthly returns are calculated from the index values using:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

The resulting return matrix R has 120 rows and 7 columns.

Next, the matrix product $R^\top R$ is calculated.

The sum of its seven diagonal entries is:

$$\text{SUM} = \sum_{i=1}^{7} (R^\top R)_{ii} = 7.7951907$$

Therefore, answer c is correct, since $5 <$ SUM < 10.

R Code

```r
# Exercise 4: Matrix Calculations
library(readxl)

# Read columns C-I from Sheet1
data_excel <- read_excel("01_data/raw/HA2_data_adj.xlsx",
                   sheet = "Sheet1",
                   range = "C1:I122")

# Convert index values to a matrix
prices <- as.matrix(data_excel)

# Calculate continuously compounded returns
R <- diff(log(prices))

# Calculate R transpose times R
RtR <- crossprod(R)

# Sum diagonal entries
sum_diagonal <- sum(diag(RtR))

# Results
sum_diagonal
```

Exercise 5: Returns and Volatility

Using the return matrix R from Exercise 4, the mean return and volatility are calculated for each of the seven assets.

The mean return is calculated as:

$$\overline{r}_i = \frac{1}{120} \sum_{t=1}^{120} r_{t,i}$$

Volatility:

$$\sigma_i = \sqrt{\frac{1}{119} \sum_{t=1}^{120} (r_{t,i} - \overline{r}_i)^2}$$

Sum of all mean returns and volatilities:

$$\text{SUM} = \sum_{i=1}^{7} (\overline{r}_i + \sigma_i) = 0.6364081$$

Therefore, answer c is correct, since $0.5 <$ SUM < 1.

R Code

```r
# Exercise 5: Returns and Volatility

# Mean return of each asset
mean_returns <- colMeans(R)

# Volatility of each asset
volatility <- apply(R, 2, sd)

# Sum of all 14 values
sum_all_values <- sum(mean_returns) + sum(volatility)

# Results
mean_returns
volatility
sum_all_values
```

Exercise 6: Quadratic Utility

Quadratic utility function:

$$U(W) = W - \frac{\lambda}{2}W^2, \quad \lambda > 0$$

Absolute Risk Aversion (ARA)

First and second derivatives:

$$U'(W) = 1 - \lambda W$$

$$U''(W) = -\lambda$$

Arrow-Pratt measure of absolute risk aversion:

$$ARA(W) = -\frac{U''(W)}{U'(W)} = \frac{\lambda}{1 - \lambda W}$$

Utility is increasing when $U'(W) > 0$, which requires $W < 1/\lambda$.

Increasing Absolute Risk Aversion (IARA)

Differentiating ARA with respect to wealth:

$$ARA'(W) = \frac{\lambda^2}{(1 - \lambda W)^2} > 0$$

Therefore, absolute risk aversion increases with wealth in the region where utility is increasing.

Therefore, answer a is correct, as both parts have been proven.

Exercise 7: Fama-French Factors

The monthly Fama-French three-factor data is obtained from the Kenneth R. French Data Library.

The dataset contains four variables: Mkt-RF, SMB, HML, and RF.

First, the monthly returns are converted from percentages into decimal values.

Next, the mean, standard deviation, and variance are calculated for each of the four variables using R.

Then the variance-covariance matrix is calculated.

The sum of all 16 entries of the covariance matrix is:

$$\text{SUM} = \sum_{i=1}^{4} \sum_{j=1}^{4} \Sigma_{ij} = 0.0072308$$

Therefore, answer b) is correct, since $0 <$ SUM < 0.5.

R Code

```r
# Exercise 7: Fama-French Factors

data_ff_raw <- read.csv(
  "01_data/raw/F-F_Research_Data_Factors.csv",
  skip = 4
)

head(data_ff_raw)


# Keep only monthly observations (YYYYMM)
data_ff_m <- data_ff_raw[
  grepl("^[0-9]{6}$", data_ff_raw$X),
]

# Convert factor columns to numeric values
data_ff_m[, 2:5] <- lapply(data_ff_m[, 2:5], as.numeric)

# Convert percent to decimals
data_ff_m[, 2:5] <- data_ff_m[, 2:5] / 100

# Check results
head(data_ff_m)
tail(data_ff_m)
nrow(data_ff_m)

# Select the four monthly return series
ff_returns <- data_ff_m[, 2:5]

# Mean, standard deviation and variance
ff_mean <- colMeans(ff_returns)
ff_sd <- sapply(ff_returns, sd)
ff_var <- sapply(ff_returns, var)
```

```r
# Variance-covariance matrix
ff_cov <- cov(ff_returns)

# Sum of all 16 matrix entries
sum_all_matrix <- sum(ff_cov)

# Results
ff_mean
ff_sd
ff_var
ff_cov
sum_all_matrix
```